

Mass spectrometry (HL)

Higher level (HL)

Learning outcomes

At the end of this section you should be able to deduce information about the structural features of a compound from specific MS fragmentation patterns.

'You are what you eat' is a phrase often used to imply that the food that you eat can have a significant effect on your health. Therefore, it is important that our food does not contain harmful chemicals. The detection of such chemicals can be carried out using a mass spectrometer. Mass spectrometry is also used in drug detection and analysis, which is especially important in sports, where athletes can get an unfair advantage through the use of performance-enhancing drugs.

Mass spectrometry was first encountered in [section S1.2.2](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/isotopes-id-43076/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/isotopes-id-43076/>) where a mass spectrum was used to calculate the relative atomic mass of an element. Furthermore, you discovered in [section S1.4.4](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/empirical-formula-and-molecular-formula-id-43572/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/empirical-formula-and-molecular-formula-id-43572/>) that this technique can be used to determine the molecular formula of a compound. In this section, we will use this technique to determine the structural features of an organic compound.

The stages in a mass spectrum are shown in **Figure 1**.

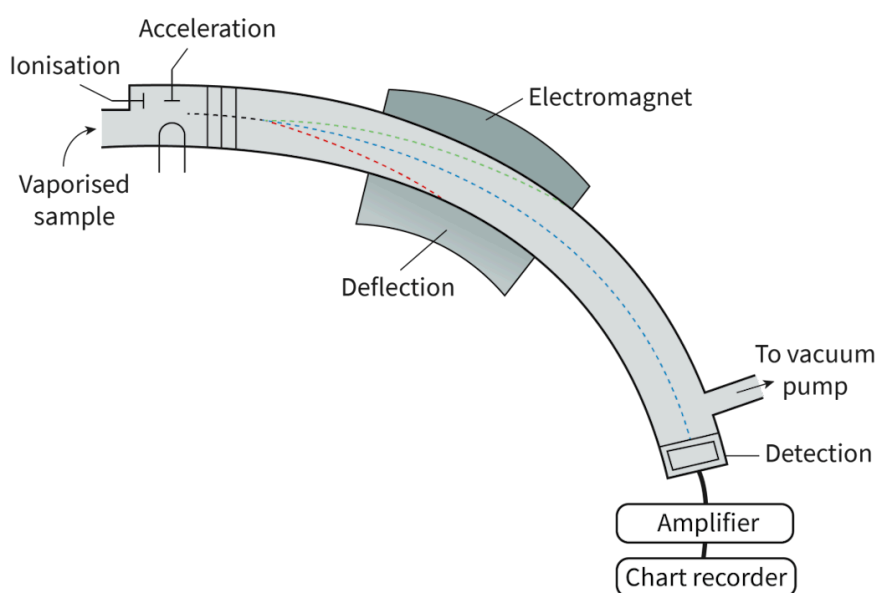


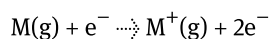
Figure 1. A mass spectrometer.

 More information for figure 1

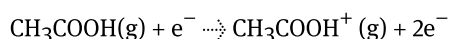
The image is a diagram illustrating the stages of a mass spectrometer. The process starts with a vaporized sample entering the ionization chamber, labeled 'Ionisation'. Here, the sample is bombarded to form ions. The ions are then accelerated through the 'Acceleration' stage, which propels them towards the electromagnet. The path bends under the influence of the 'Electromagnet', causing 'Deflection'. The deflected ions are then detected in the 'Detection' stage, which is connected to a 'To vacuum pump'. The process culminates with signals being sent to an 'Amplifier' and then to a 'Chart recorder', which logs the results. Each stage is clearly labeled in the diagram, illustrating the flow of the sample through the various components.

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We will now focus on the ionisation stage. In the ionisation stage, the gaseous compound is bombarded with high energy electrons, forming a positive ion known as the molecular ion. The molecular ion is represented as M^+ or $M\cdot^+$. The dot represents a single unpaired electron that is left over when an electron is lost from a pair of electrons somewhere in the molecule. The ionisation process can be represented in equation form as follows where M is the molecule in question and M^+ the molecular ion.



For the compound ethanoic acid, the equation would be:



Note that the formula of the molecular ion is the same as that of the compound except that it has a positive charge as a result of losing an electron. The molecular ion itself is unstable and can break into pieces which are known as fragments. This involves the breaking of covalent bonds between the atoms in the molecule. The ion breaks into two fragments: one positively charged ion (the cation fragment) and one neutral fragment. This is summarised in **Figure 2**. The molecular ion can break in different places within the ion, which leads to many different possible cation fragments.



Figure 2. The molecular ion is broken up (fragmented) into a cation fragment and neutral fragment.

 More information for figure 2

The diagram illustrates the fragmentation of a molecular ion, represented as ' $M\cdot^+$ ', into two parts as a result of the fragmentation process. There is a horizontal arrow indicating the direction of fragmentation from left to right. The molecular ion ' $M\cdot^+$ ' breaks into a 'cation fragment', labeled as ' m^+ ', and a 'neutral fragment', labeled as ' $m\cdot$ '. The positions of these fragments are indicated with text labels beneath them, specifying 'Molecular ion' for the initial molecule, and 'Cation fragment' and 'Neutral fragment' for the resulting fragments. The word 'Fragmentation' is shown above the arrow.

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Only the cation fragments are detected in the mass spectrometer and produce a fragmentation pattern. The neutral fragments are not detected and are removed from the mass spectrometer. By analysing a mass spectrum that shows the fragmentation pattern produced we can determine certain structural features of the compound.

Figure 3 shows the mass spectrum and fragmentation pattern of ethanoic acid, CH_3COOH . The molecular ion (M^+) peak can be identified on the far right of the spectrum with a m/z of 60 (this is the peak with highest m/z on the spectrum). A quick calculation shows that this is approximately the same as the molar mass of ethanoic acid, which is 60.06 g mol^{-1} . From the m/z of the molecular ion peak, we can determine the molar mass, M (or relative molecular mass, M_r), of the compound.

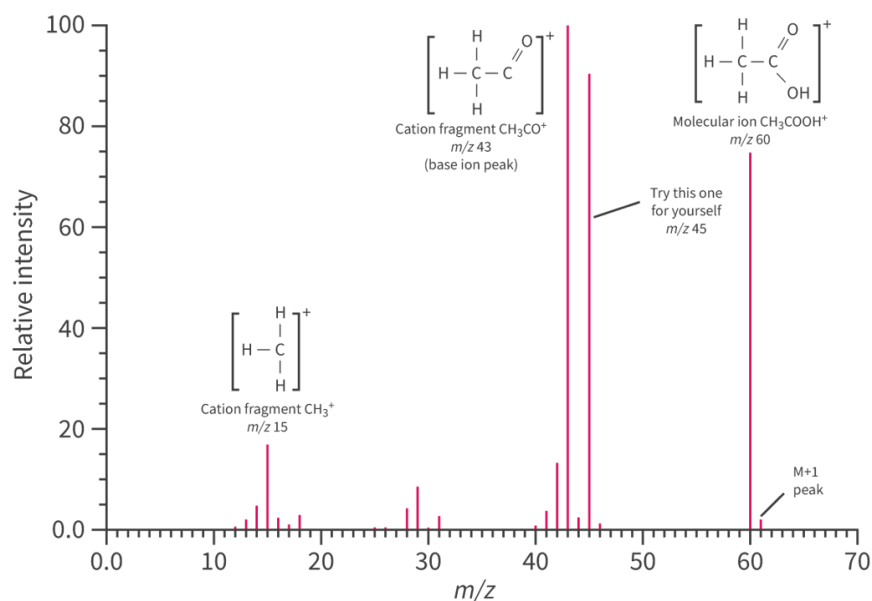


Figure 3. Mass spectrum showing the fragmentation pattern of ethanoic acid.

 More information for figure 3

The image displays a mass spectrum graph with the X-axis labeled as ' m/z ' ranging from 0 to 70, and the Y-axis labeled as 'Relative intensity' with values from 0 to 100. The main peaks are identified at different m/z values: 15, 43, and 60. The peak at m/z 60 is labeled as the molecular ion peak for CH_3COOH^+ , indicating the presence of the entire ethanoic acid molecule. The peak at m/z 43 is labeled as the base ion peak for the CH_3CO^+ fragment. Another peak at m/z 15 represents the CH_3^+ cation fragment. There is also a reference to an m/z 45 peak labeled 'Try this one for yourself.' A smaller peak labeled 'M+1' appears at m/z 61. This spectrum illustrates how ethanoic acid fragments in a mass spectrometer, showing the relative abundance of each ionized fragment.

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Study skills

Next to the molecular ion peak, another much smaller peak can be seen. This is the $M+1$ peak and is produced by the presence of the isotopes carbon-13 or hydrogen-2 in the compound. Note that this peak is often omitted in simplified mass spectra such as those used in IB exams.

It is important to note that all the peaks shown on a mass spectrum are produced by positively charged fragments because the neutral fragments are not detected. Because they all have a $1+$ charge, the m/z ratio of the fragment is the same as its mass. So, if a cation fragment has a m/z of 15, its mass is also 15.

The fragmentation pattern is analysed in conjunction with **Table 1** (from section 22 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/mass-spectral-fragments-lost-id-45124/>) of the DP chemistry data booklet). **Table 1** shows the mass lost from the molecular ion and the possible neutral fragment lost. So, a mass of 15 lost from the molecular ion could possibly correspond to the loss of a methyl radical, $\bullet\text{CH}_3$; a mass of 17 lost from the molecular ion could possibly correspond to the loss of an $\bullet\text{OH}$ radical and so on.

Table 1. The mass lost from the molecular ion and the possible neutral fragment lost (reproduced from section 22 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/mass-spectral-fragments-lost-id-45124/>) of the DP chemistry data booklet).

Mass lost (M_r)	Possible neutral fragment lost
15	$\bullet\text{CH}_3$
17	$\bullet\text{OH}$
18	H_2O
28	$\text{CH}_2=\text{CH}_2\text{CO}$
29	$\bullet\text{CH}_2\text{CH}_3$ $\bullet\text{CHO}$
31	$\bullet\text{OCH}_3$
45	$\bullet\text{COOH}$

Now we will look at the relevance of the other peaks on the spectrum, specifically those at m/z 43 and m/z 15. Starting with the peak at m/z 43, we will determine the mass lost and the possible neutral fragment lost. This is done by subtracting the mass of the cation fragment from the mass of the molecular ion which gives the mass of the neutral fragment lost. This can be summarised as:

$$m/z \text{ of molecular ion} - m/z \text{ of cation fragment} = \text{mass of neutral fragment lost}$$

For the peak at m/z 43, this would be:

$$\begin{aligned} &\text{mass of molecular ion (60)} - \text{mass of cation fragment (43)} = \\ &\text{mass of neutral fragment lost (17)} \end{aligned}$$

Looking at Table 1, we see that this could possibly correspond to the loss of the neutral fragment $\bullet\text{OH}$. If we remove an OH from the molecular ion, we are left with the cation fragment CH_3CO^+ , which has a mass of 43 and produces the peak at m/z 43. **Figure 4** summarises this process in which fragmentation of the molecular ion CH_3COOH^+ produces the cation fragment CH_3CO^+ and the neutral fragment $\bullet\text{OH}$.

Looking back at the mass spectrum in **Figure 3**, we can see that the cation fragment CH_3CO^+ has a relative intensity of 100% and is known as the base ion peak. The base ion peak represents the most abundant cation fragment detected in the mass spectrometer.

Study skills

When calculating the mass of the molecular ion, cation fragment or the mass of the neutral fragment lost, use whole-number molar masses such as 12 for carbon, 16 for oxygen and 1 for hydrogen.

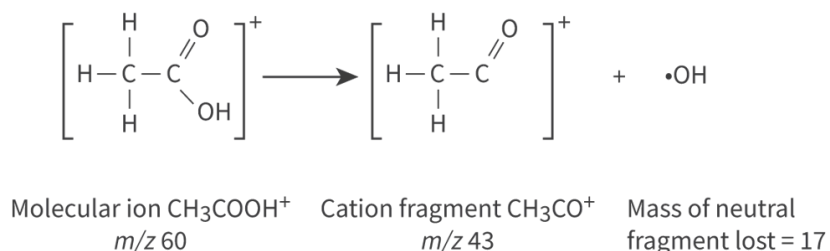


Figure 4. Fragmentation of CH_3COOH^+ resulting in a mass lost of 17.

More information for figure 4

The image is a chemical diagram showing the fragmentation of the CH_3COOH^+ molecular ion. On the left side, the molecular ion CH_3COOH^+ is depicted with a structural formula indicating the arrangement of atoms: carbon (C) atoms and hydrogen (H) atoms bonded together with an oxygen (O) atom double-bonded to a carbon, and an OH group attached. It is labeled with a mass-to-charge ratio (m/z) of 60. To the right, an arrow indicates the process of fragmentation resulting in a cation fragment, CH_3CO^+ , also shown with its structural formula and labeled with an m/z of 43. A hydroxyl radical, $\bullet\text{OH}$, is shown as a product. Below the reaction, it states "Mass of neutral fragment lost = 17."

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Next we will look at the peak at m/z 15. Using the same equation as in the previous example we get:

$$\text{mass of molecular ion (60)} - \text{mass of cation fragment (15)} = \text{mass of neutral fragment lost (45)}$$

Looking at **Table 1**, this could possibly correspond to the loss of a $\bullet\text{COOH}$ radical, which has a mass of 45. The fragment that caused the peak at m/z has a mass of 15 and this corresponds to the cation fragment CH_3^+ . **Figure 5** summarises the fragmentation of the molecular ion CH_3COOH^+ to produce the cation fragment CH_3^+ and the neutral fragment $\bullet\text{COOH}$.

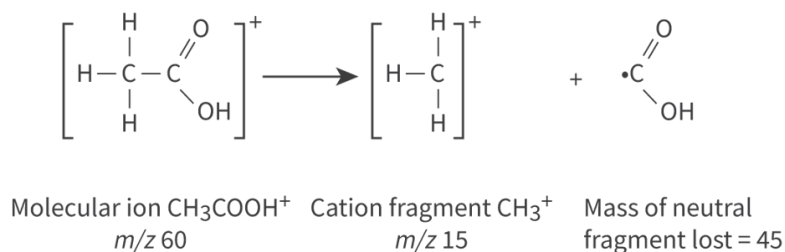


Figure 5. Fragmentation of CH_3COOH^+ resulting in a mass lost of 45.

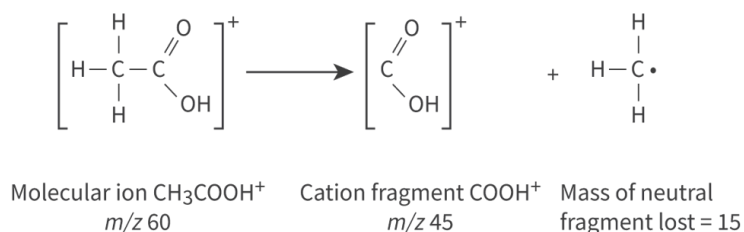
 More information for figure 5

The diagram illustrates the fragmentation process of the molecular ion CH_3COOH^+ into two parts: the cation fragment CH_3^+ and the neutral fragment COOH . On the left side, the structure of the acetic acid ion CH_3COOH^+ is shown, labeled with a mass-to-charge ratio (m/z) of 60. An arrow points to the right side, indicating the fragmentation process. To the right of the arrow, the cation fragment CH_3^+ is displayed, labeled with an m/z of 15. Adjacent to it, the neutral fragment COOH , which is missing one electron, is depicted. Below the COOH fragment, there is a label indicating that the mass of the neutral fragment lost is 45. The image effectively demonstrates the breaking of the bond in the acetic acid ion to form separate ionic and neutral fragments.

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And finally, we will look at the peak at m/z 45. Try to see whether you can work out the mass lost and the possible neutral fragment lost for yourself. Check your answer when you are ready and don't forget to refer to **Table 1**.

The mass lost is 15, which possibly corresponds to loss of a $\bullet\text{CH}_3$ radical. The peak at m/z 45 was produced by the cation fragment COOH^+ .



To summarise, by analysing the fragmentation pattern produced in a mass spectrometer, we can determine the mass lost from the molecular ion. This gives us clues as to the structural features of the molecule in question and helps us to identify the unknown organic compound.

Worked example 1

The mass spectrum below is for a compound with the empirical formula $\text{C}_2\text{H}_6\text{O}$.

Use the mass spectrum and table x to determine the following:

1. the relative molecular mass, M_r , of the compound
2. the m/z of the base ion peak
3. the formula of the molecular ion
4. the mass lost and possible neutral fragment lost for the fragment at m/z 15 and 31.

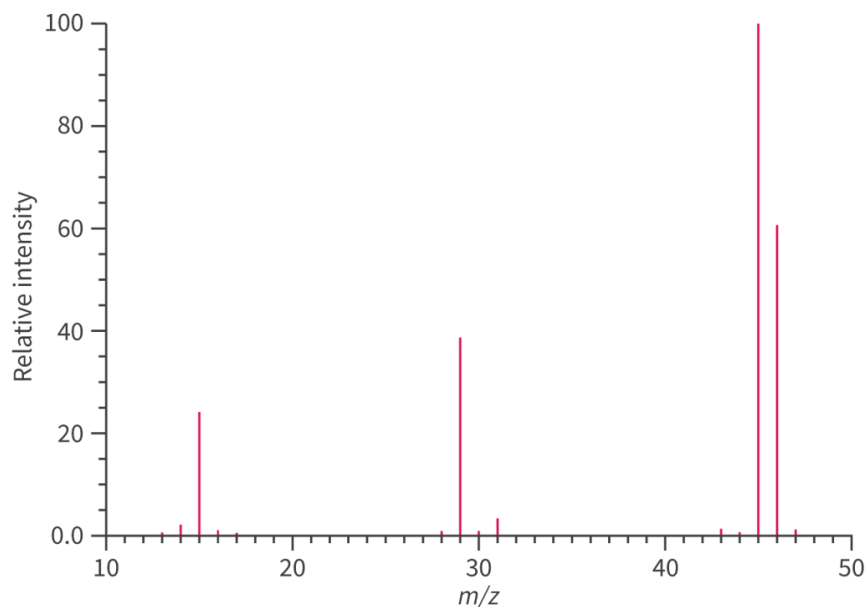


Figure 6. The mass spectrum for a compound with the empirical formula C_2H_6O .

 More information for figure 6

This is a mass spectrum graph for a compound with the empirical formula C_2H_6O . The X-axis represents the mass-to-charge ratio (m/z) and ranges from 10 to 50. The Y-axis represents the relative intensity and ranges from 0 to 100. There are prominent peaks at m/z values 15, 30, and 45, with the highest intensity peak at m/z 45. Smaller peaks are visible at m/z values 18 and 31, indicating fragment ions of the compound.

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1. The molecular ion peak has a m/z of 46 so this is the relative molecular mass.
2. The base ion peak has an m/z of 45.
3. $C_2H_6O^+$
4. m/z 15: mass lost: $46 - 15 = 31$
Possible neutral fragment lost is $\bullet OCH_3$
 m/z 31: mass lost $46 - 31 = 15$
Possible neutral fragment lost is $\bullet CH_3$

In this activity, you will interpret two mass spectra to determine the structural features of two organic compounds.

Activity

- **IB learner profile attribute:** Knowledgeable
- **Approaches to learning:** Thinking skills — Applying key ideas and facts in new contexts
- **Time required to complete activity:** 30 minutes
- **Activity type:** Individual activity

1. The mass spectrum of an unknown organic compound is shown in **Figure 7**. Study the mass spectrum and answer the following questions.

- Determine the molar mass of the compound.
- Deduce the m/z of the most stable ion.
- The mass lost and possible neutral fragment lost for the fragment at m/z 31.
- The mass lost and possible neutral fragment lost for the fragment at m/z 29.

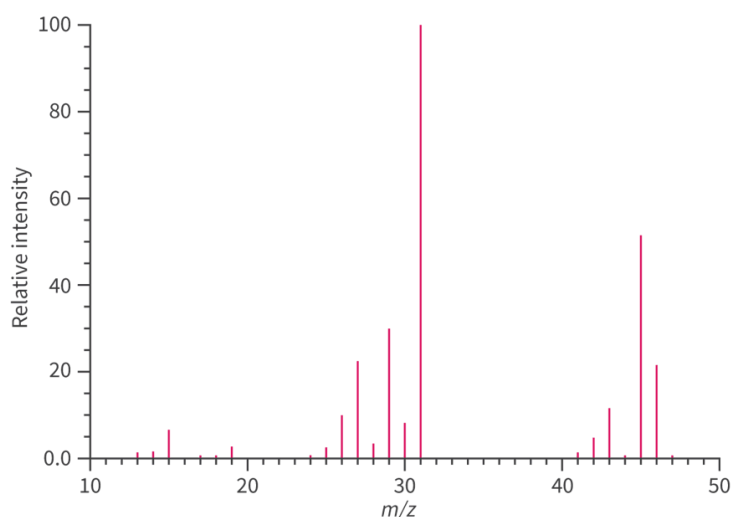


Figure 7. The mass spectrum of an unknown organic compound.

 More information for figure 7

The image is a mass spectrum graph of an unknown organic compound. The X-axis represents the mass-to-charge ratio (m/z), ranging from 10 to 50. The Y-axis indicates the relative intensity, ranging from 0 to 100. Various peaks are present in the spectrum. Notably, there is a prominent peak at around m/z 30 with the highest relative intensity close to 100. Other significant peaks are observed at approximately m/z 15, 28, and 44, with varying relative intensities. These peaks represent different fragments of the compound and are useful for deducing the molecular structure, identifying the molar mass, and determining the most stable ion of the compound.

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2. The mass spectrum of an unknown organic compound is shown in **Figure 8**. Study the mass spectrum and answer the following questions.

- Determine the molar mass of the compound.
- Deduce the m/z of the most stable ion.
- The mass lost and possible neutral fragment lost for the fragment at m/z 29.
- The mass lost and possible neutral fragment lost for the fragment at m/z 15.

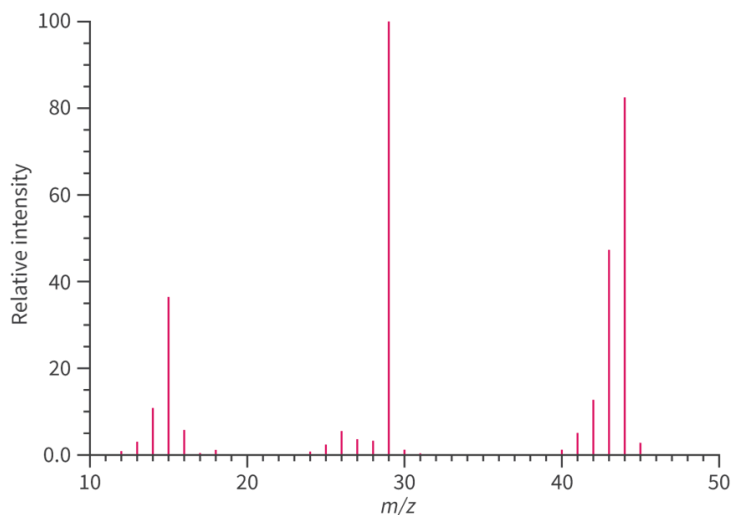


Figure 8. The mass spectrum of an unknown organic compound.

 More information for figure 8

The image is a mass spectrum graph of an unknown organic compound. The X-axis represents the m/z (mass-to-charge ratio) values ranging from 10 to 50, while the Y-axis represents the relative intensity, scaling from 0 to 100. The spectrum shows multiple peaks with the most prominent peak at m/z 32, suggesting this is the base peak with the highest relative intensity of 100. There are smaller peaks at m/z values like 15, 28, and other mid-range values, indicating the presence of other ions or fragments. This graph helps determine the molar mass, stability, and potential fragment losses of the compound based on the peaks and their positions.

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- Approximately 46 g mol^{-1}
 - The most stable ion is the base ion peak that has a m/z of 31
 - Mass lost: $46 - 31 = 15$ neutral fragment lost is $\bullet\text{CH}_3$
 - Mass lost: $46 - 29 = 17$ neutral fragment lost is $\bullet\text{OH}$
- Approximately 44 g mol^{-1}
 - The most stable ion is the base ion peak that has a m/z of 29
 - Mass lost: $44 - 29 = 15$ neutral fragment lost is $\bullet\text{CH}_3$
 - Mass lost: $44 - 15 = 29$ neutral fragment lost is $\bullet\text{CHO}$

